

Xinyu Zhang

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INTRODUCTION

I am a CS PhD candidate (5th year, graduating this August) at **Rutgers University**, focusing on **bringing 2D/3D world understanding to robotics policy**. I published 7 first-author papers at top robotics and vision venues. I worked as a research intern at **Meta Reality Lab** and an ML engineer at **Microsoft**. My publications span robot manipulation policy learning, world modeling, computer vision, 3D reconstruction, and video generation.

PUBLICATION

Xinyu Zhang, Zhengtong Xu, Yutian Tao, Yeping Wang, Yu She, Abdeslam Boularias.

Learning Visual Feature-based World Models via Residual Latent Action

NeurIPS 2026 Under Review [\[website\]](#) [\[code\]](#) [\[arxiv\]](#)

- Proposed **RLA-WM**, a simple and efficient state-of-the-art world model.
- The first demonstration of visual reinforcement learning inside a world model that learned from only offline videos, without online rollouts or handcrafted rewards.

Xinyu Zhang, Meta Reality Lab.

Glove2Hand: Synthesizing Natural Hand-Object Interaction from Multi-Modal Sensing Gloves.

CVPR 2026 Highlight [\[website\]](#) [\[arxiv\]](#)

- Proposed **Glove2Hand**, a 3d reconstruction + video generation framework that synthesizes temporally consistent, photorealistic real-hand object interaction from sensor gloves.
- It enables force sensing in human videos and aligns videos of multiple hand manipulators.

Xinyu Zhang, Haonan Chang, Yuhan Liu, Abdeslam Boularias.

Motion Blender Gaussian Splatting for Dynamic Scene Reconstruction.

CoRL 2025 [\[code & demo\]](#) [\[arxiv\]](#)

- Proposed **Motion-Blender-GS**, which reconstructs dynamic scenes into Gaussian splats that are controlled by motion graphs. The motion graph is an explicit and sparse motion representation, which enables digital twin and real2sim2real after reconstruction

Xinyu Zhang, Yuhan Liu, Haonan Chang, Liam Schramm, Abdeslam Boularias.

Autoregressive Action Sequence Learning for Robotic Manipulation.

RAL 2025 [\[code & demo\]](#) [\[arxiv\]](#)

- Proposed **Autoregressive Policy**, a unified LLM architecture for policy learning that uses variable-size chunking to support action prediction during token generation. This unified framework established SOTA in multiple environments at the time

Xinyu Zhang, Yuhan Liu, Yuting Wang, Abdeslam Boularias.

Detect Everything with Few Examples.

CoRL 2024 [\[code\]](#) [\[demo\]](#) [\[arxiv\]](#)

- Proposed **DE-ViT**, the first example-driven open-world detector that detects new categories instantly without any tuning
- Achieved SOTA on LVIS, COCO, and outperforms the second by +10 mAP at the time

Xinyu Zhang, Abdeslam Boularias.

One-Shot Imitation Learning with Invariance Matching for Robotic Manipulation.

RSS 2024 [\[code\]](#) [\[video\]](#) [\[arxiv\]](#)

- Proposed **IMOP**, a one-shot policy, which solves novel quasi-static manipulation tasks by matching 3D regions between demonstrations and testing scenes, and transferring

end-effector poses through matched correspondence

- The policy delivered non-trivial success rates (41%, +11% compared to retraining baseline) on 22 novel tasks on RL Bench, with just a single demonstration per task

Xinyu Zhang, Yuhan Liu, Haonan Chang, Abdeslam Boularias.

Scaling Manipulation Learning with Visual Kinematic Chain Prediction.

CoRL 2024 [\[code\]](#) [\[demo\]](#) [\[arxiv\]](#)

- Proposed VKT, a generalist policy that effectively bridges embodiment and workspace gaps by forecasting the movements of the visual kinematic chain as a universal action space (e.g., from 10% to 55% success rate for a generalist on RL Bench).

Yuhan Liu, **Xinyu Zhang**, Haonan Chang, and Abdeslam Boularias.

Failure Forecasting Boosts Robustness of Sim2Real Rhythmic Insertion Policies.

IROS 2025 [\[code\]](#) [\[demo\]](#) [\[arxiv\]](#)

- Proposed an online estimation model that learns to predict policy success rates, which solves repeated insertion tasks with sim-to-real transfer

Haonan Chang, Kowndinya Boyalakuntla, Yuhan Liu, **Xinyu Zhang**, Liam Schramm, Abdeslam Boularias.

Diffusion-based Affordance Prediction for Multi-modality Storage.

IROS 2024 [\[code\]](#) [\[arxiv\]](#)

- Proposed a diffusion policy that predicts an affordance matching (similar to the region matching in IMOP) from multiple storage slots. The insertion pose is derived from the matched regions

Xinyu Zhang, Abdeslam Boularias.

Optical Flow boosts Unsupervised Localization and Segmentation.

IROS 2023 [\[code\]](#) [\[arxiv\]](#)

- Proposed Flow-DINO, which uses optical flow as a self-supervised learning objective, to encourage object-aware spatial features. The learned features improve downstream object localization and segmentation performance

Xinyu Zhang, Ian Colbert, Srinjoy Das.

Learning Low-Precision Structured Subnetworks Using Joint Layerwise Channel Pruning and Uniform Quantization.

MDPI 2022 [\[code\]](#) [\[paper\]](#)

- Proposed Recursive Pruning, which identifies inter-layer neuron dependencies, and prunes layer by layer in topological order, outperforms magnitude-based pruning consistently over classification, segmentation, and image generation

Xinyu Zhang, Srinjoy Das, Ken Kreutz-Delgado.

Tuning Confidence Bound for Stochastic Bandits with Bandit Distance.

Arxiv 2021 [\[paper\]](#)

- Proposed UCB-DT, which is a stochastic bandit method informed by estimating the distribution distance between the rewards of different arms, with better theoretical bound than UCB, and consistently outperforms other UCB variants such as UCB-dagger

Mengting Chen, Yuxin Fang, Xinggang Wang, Heng Luo, Yifeng Geng, **Xinyu Zhang**, Chang Huang, Wenyu Liu, Bo Wang.

Diversity transfer network for few-shot learning.

AAAI 2020 [\[paper\]](#)

- Classify any objects by generating features from example images

EDUCATION

Rutgers University PhD Candidate, Computer Science	2022 - Present
University of California, San Diego M.S., Computer Science	2015 - 2017
University of Science and Technology of China B.E., Electrical Engineering	2011 - 2015

INDUSTRIAL EXPERIENCE

Meta Research Scientist Intern	2025 Jun - Dec
- Proposed Glove2Hand, a 3D-2D hybrid <u>generative</u> framework that integrates reconstruction and video diffusion for consistent and flexible hand generation. - My proposed method reconstructs photorealistic bare-hand video from tactile glove videos, which enables <u>multi-modal sensing</u> with egocentric bare hands for object manipulation	
Microsoft Software Engineer II	2020 - 2022
Developed <u>contextual bandit</u> methods such as LinUCB and Thompson Sampling, for reranking ads, news, and widgets in the Edge browser, using C#, Kafka, and Flink	
Megvii (Face++) Machine Learning Engineer	2018 - 2020
Developed large-scale face recognition networks, based on <u>representation learning</u> with contrastive loss, AMSoftmax, and self-training over billion-scale datasets	

SKILLS

- Python, PyTorch, JavaScript, C/C++, CUDA, ROS 1, LaTeX, Slurm / Kubernetes, MuJoCo, SQL, MoCap System, OptiTrack, MANO, SMPL
- Imitation Learning, Model Compression, Reinforcement Learning, Self-Supervised Learning, Open-Vocabulary Detection/Segmentation
- Point Transformer, Spatially Sparse CNN, Point Cloud Registration, DINOv2, DETR, CLIP, MaskFormer, ViT, UNet, Mask-RCNN, Diffusion, VAE, GAN, ControlNet